

Q1

```
SQLQuery4.s...d Raza (800)*
1  -- Q1
2  BEGIN TRANSACTION;
3  BEGIN TRY
4      INSERT INTO match VALUES (206, '2025-04-01', 'Multan', 1, 4, 1, 2);
5      INSERT INTO performance VALUES (101, 206, 45, 0, 1);
6      INSERT INTO performance VALUES (102, 206, 5, 2, 0);
7      COMMIT;
8      PRINT 'Committed.';
9  END TRY
10 BEGIN CATCH
11     ROLLBACK;
12     PRINT 'Rolled back: ' + ERROR_MESSAGE();
13 END CATCH
14
15 -- Q2
16 BEGIN TRANSACTION;
17 BEGIN TRY
18     UPDATE player SET tid = 2 WHERE pid = 105;
19     UPDATE player SET captain id = NULL WHERE captain id = 105;
20
132 % 132 % No issues found Ln: 1, Ch: 1 (358 chars, 13 lines) SPC CRLF
Messages
(1 row affected)
(1 row affected)
(1 row affected)
Committed.
Completion time: 2026-04-26T11:16:21.1030979-07:00
```

After trying the same query again.

```
SQLQuery4.s...d Raza (800)*
1  -- Q1
2  BEGIN TRANSACTION;
3  BEGIN TRY
4      INSERT INTO match VALUES (206, '2025-04-01', 'Multan', 1, 4, 1, 2);
5      INSERT INTO performance VALUES (101, 206, 45, 0, 1);
6      INSERT INTO performance VALUES (102, 206, 5, 2, 0);
7      COMMIT;
8      PRINT 'Committed.';
9  END TRY
10 BEGIN CATCH
11     ROLLBACK;
12     PRINT 'Rolled back: ' + ERROR_MESSAGE();
13 END CATCH
14
15 -- Q2
16 BEGIN TRANSACTION;
17 BEGIN TRY
18     UPDATE player SET tid = 2 WHERE pid = 105;
19     UPDATE player SET captain id = NULL WHERE captain id = 105;
20
132 % 132 % No issues found Ln: 1, Ch: 1 (356 chars, 13 lines) SPC CRLF
Messages
(0 rows affected)
Rolled back: Violation of PRIMARY KEY constraint 'PK_match_DF5032ECB3AE4BD6'. Cannot insert duplicate key in object 'dbo.match'. The duplicate key value is (206).
Completion time: 2026-04-26T11:27:06.9020011-07:00
```

Q2

```
SQLQuery4.s...d Raza (80) |
10 BEGIN CATCH
11     ROLLBACK;
12     PRINT 'Rolled back: ' + ERROR_MESSAGE();
13 END CATCH
14
15 -- Q2
16 BEGIN TRANSACTION;
17 BEGIN TRY
18     UPDATE player SET tid = 2 WHERE pid = 105;
19     UPDATE player SET captain_id = NULL WHERE captain_id = 105;
20 COMMIT;
21 PRINT 'Committed.';
22 END TRY
23 BEGIN CATCH
24     ROLLBACK;
25     PRINT 'Rolled back: ' + ERROR_MESSAGE();
26 END CATCH
27
28 -- Q3
132 % | No issues found Ln: 15, Ch: 1 (281 chars, 12 lines) SPC CRLF U
Messages
(1 row affected)
(1 row affected)
Committed.
Completion time: 2026-04-26T11:28:06.9411374-07:00
```

Q3

```
SQLQuery4.s...d Raza (80) |
25 PRINT 'Rolled back: ' + ERROR_MESSAGE();
26 END CATCH
27
28 -- Q3
29 BEGIN TRANSACTION;
30 DECLARE @t1 INT = 1, @t2 INT = 4;
31 IF EXISTS (SELECT 1 FROM team WHERE tid = @t1)
32 AND EXISTS (SELECT 1 FROM team WHERE tid = @t2)
33 BEGIN
34     INSERT INTO match VALUES (207, '2025-05-01', 'Lahore', @t1, @t2, @t1, 2);
35 COMMIT;
36 PRINT 'Committed.';
37 END
38 ELSE
39 BEGIN
40     ROLLBACK;
41     PRINT 'Rolled back: one or both teams do not exist.';
42 END
43
44 -- Q4
120 % | No issues found Ln: 28, Ch: 1 (381 chars, 15 lines) SPC CRLF I
Messages
(1 row affected)
Committed.
Completion time: 2026-04-26T11:28:56.1899730-07:00
```

Q4

```
DESKTOP-IK3...performance Solution.sql...ad Raza (800)
41 PRINT 'Rolled back: one or both teams do not exist.';
42 END
43
44 -- Q4
45 BEGIN TRANSACTION;
46 BEGIN TRY
47     INSERT INTO tournament VALUES (11, 'Asjads League', 2026);
48     SAVE TRANSACTION sp1;
49     INSERT INTO match VALUES (225, '2026-01-01', 'Karachi', 1, 2, 1, 11);
50     INSERT INTO match VALUES (226, '2026-01-05', 'Lahore', 3, 4, 3, 11);
51     COMMIT;
52     PRINT 'Committed: tournament and matches saved.';
53 END TRY
54 BEGIN CATCH
55     ROLLBACK TRANSACTION sp1;
56     PRINT 'Rolled back to savepoint: invalid teams in matches.';
57     COMMIT;
58     PRINT 'Committed: tournament saved.';
59 END CATCH
60
61 -- Q5
62 BEGIN TRANSACTION;
63 UPDATE player SET salary = salary * 1.10 WHERE tid = 1;
64 IF (SELECT AVG(salary) FROM player WHERE tid = 1) > 90000
65 BEGIN
66     ROLLBACK TRANSACTION;
67     PRINT 'Salary increase failed: average salary too high.';
68     COMMIT;
69     PRINT 'Committed: salary update failed.';
70 END
71 COMMIT;
72 PRINT 'Committed: salary update successful.';
73 END TRANSACTION;
```

90 % No issues found Ln: 45, Ch: 1 (530 chars, 16 lines) SPC CRLF UTF

Messages

(1 row affected)
(1 row affected)
(1 row affected)
Committed: tournament and matches saved.
Completion time: 2026-04-26T11:56:48.8418508-07:00

```
DESKTOP-IK3...performance Solution.sql...ad Raza (800)
43
44 -- Q4
45 BEGIN TRANSACTION;
46 BEGIN TRY
47     INSERT INTO tournament VALUES (10, 'Asjads League', 2026);
48     SAVE TRANSACTION sp1;
49     INSERT INTO match VALUES (220, '2026-01-01', 'Karachi', 1, 2, 1, 10);
50     INSERT INTO match VALUES (221, '2026-01-05', 'Lahore', 99, 4, 3, 10);
51     COMMIT;
52     PRINT 'Committed: tournament and matches saved.';
53 END TRY
54 BEGIN CATCH
55     ROLLBACK TRANSACTION sp1;
56     PRINT 'Rolled back to savepoint: invalid teams in matches.';
57     COMMIT;
58     PRINT 'Committed: tournament saved.';
59 END CATCH
60
61 -- Q5
62 BEGIN TRANSACTION;
63 UPDATE player SET salary = salary * 1.10 WHERE tid = 1;
64 IF (SELECT AVG(salary) FROM player WHERE tid = 1) > 90000
65 BEGIN
66     ROLLBACK TRANSACTION;
67     PRINT 'Salary increase failed: average salary too high.';
68     COMMIT;
69     PRINT 'Committed: salary update failed.';
70 END
71 COMMIT;
72 PRINT 'Committed: salary update successful.';
73 END TRANSACTION;
```

90 % No issues found Ln: 45, Ch: 1 (531 chars, 16 lines) SPC CRLF UTF

Messages

(1 row affected)
(1 row affected)
(0 rows affected)
Rolled back to savepoint: invalid teams in matches.
Committed: tournament saved.
Completion time: 2026-04-26T11:54:28.1241272-07:00

Q5



The screenshot displays a SQL Server Enterprise Manager window with a T-SQL script editor and a Messages pane. The script, located in a file named 'Solution.sql...ad Raza (80)', is a transactional query that updates the salary of a player with tid = 1 by 10% and then checks if the average salary for that team exceeds 90,000. If it does, the transaction is rolled back; otherwise, it is committed. The Messages pane shows the execution results, indicating that 2 rows were affected and the transaction was rolled back because the average salary exceeded 90,000. The completion time is 2026-04-26T11:42:15.2196596-07:00.

```
65      ROLLBACK;
66      PRINT 'Rolled back: ' + ERROR_MESSAGE();
67  END CATCH
68
69  -- Q5
70  BEGIN TRANSACTION;
71  UPDATE player SET salary = salary * 1.10 WHERE tid = 1;
72  IF (SELECT AVG(salary) FROM player WHERE tid = 1) > 90000
73  BEGIN
74      ROLLBACK;
75      PRINT 'Rolled back: average salary exceeds 90000.';
76  END
77  ELSE
78  BEGIN
79      COMMIT;
80      PRINT 'Committed.';
81  END
82
```

132 % No issues found Ln: 69, Ch: 1 (281 chars, 13 lines) SPC CRLF U

Messages

(2 rows affected)
Rolled back: average salary exceeds 90000.
Completion time: 2026-04-26T11:42:15.2196596-07:00

Q6

```
DESKTOP-1K3...performance  DESKTOP-1K34...t - dbo.team  DESKTOP-1K3... - dbo.match  Solution.sql...ad Raza (80)  X  ...
81      END
82
83      -- Q6
84      BEGIN TRANSACTION;
85      INSERT INTO performance VALUES (103, 206, 90, 0, 1);
86      INSERT INTO performance VALUES (104, 206, 30, 1, 0);
87      INSERT INTO performance VALUES (107, 206, 20, 2, 0);
88      IF (SELECT SUM(runs) FROM performance WHERE mid = 206) > 500
89      BEGIN
90          ROLLBACK;
91          PRINT 'Rolled back: total runs exceed 500.';
92      END
93      ELSE
94      BEGIN
95          COMMIT;
96          PRINT 'Committed.';
97      END
98
132 %  No issues found  Ln: 83, Ch: 1 (382 chars, 15 lines)  SPC  CRLF  U
Messages
(1 row affected)
(1 row affected)
(1 row affected)
Committed.
Completion time: 2026-04-26T11:43:31.6268570-07:00
```

```
DESKTOP-1K3...performance  Solution.sql...ad Raza (80)  X  ...
81      END
82
83      -- Q6
84      BEGIN TRANSACTION;
85      INSERT INTO performance VALUES (101, 208, 510, 0, 1);
86      IF (SELECT SUM(runs) FROM performance WHERE mid = 208) > 500
87      BEGIN
88          ROLLBACK;
89          PRINT 'Rolled back: total runs exceed 500.';
90      END
91      ELSE
92      BEGIN
93          COMMIT;
94          PRINT 'Committed.';
95      END
96
97      -- Q7
98      BEGIN TRANSACTION;
99
132 %  No issues found  Ln: 83, Ch: 1 (275 chars, 13 lines)  SPC  CRLF  UT
Messages
(1 row affected)
Rolled back: total runs exceed 500.
Completion time: 2026-04-26T11:45:36.4640480-07:00
```

Q7

```
DESKTOP-1K3...performance | Solution.sql...ad Raza (800) | ...
93 COMMIT;
94 PRINT 'Committed.';
95 END
96
97 -- Q7
98 BEGIN TRANSACTION;
99 INSERT INTO team VALUES ('Sialkot Stallions', 8, 'Moin Khan', 'Sialkot');
100 SAVE TRANSACTION sp1;
101 INSERT INTO player VALUES ('Kamran Akmal', 114, 38, 'Wicketkeeper', 60000, 8, NULL);
102 INSERT INTO player VALUES ('Umar Akmal', 115, 33, 'Batsman', -5000, 8, 114);
103 INSERT INTO player VALUES ('Adnan Akmal', 116, 35, 'Batsman', 55000, 8, 114);
104 IF EXISTS (SELECT 1 FROM player WHERE tid = 8 AND salary < 0)
105 BEGIN
106     ROLLBACK TRANSACTION sp1;
107     PRINT 'Rolled back to savepoint: invalid player data.';
108 END
109 COMMIT;
110 PRINT 'Committed: team insert saved.';
```

120 % | No issues found | Ln: 97, Ch: 1 (582 chars, 14 lines) | SPC | CRLF | U

Messages

(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
Rolled back to savepoint: invalid player data.
Committed: team insert saved.
Completion time: 2026-04-26T11:46:57.5017462-07:00

```
DESKTOP-1K3...performance | Solution.sql...ad Raza (800) | ...
96
97 -- Q7
98 BEGIN TRANSACTION;
99 INSERT INTO team VALUES ('Sialkot Stallions', 8, 'Moin Khan', 'Sialkot');
100 SAVE TRANSACTION sp1;
101 INSERT INTO player VALUES ('Kamran Akmal', 114, 38, 'Wicketkeeper', 60000, 8, NULL);
102 INSERT INTO player VALUES ('Umar Akmal', 115, 33, 'Batsman', -5000, 8, 114);
103 INSERT INTO player VALUES ('Adnan Akmal', 116, 35, 'Batsman', 55000, 8, 114);
104 IF EXISTS (SELECT 1 FROM player WHERE tid = 8 AND salary < 0)
105 BEGIN
106     ROLLBACK TRANSACTION sp1;
107     PRINT 'Rolled back to savepoint: invalid player data.';
108 END
109 COMMIT;
110 PRINT 'Committed: team insert saved.';
111
112 SELECT * FROM team;
```

120 % | No issues found | Ln: 112, Ch: 1 (19 chars, 1 lines) | SPC | CRLF | U

Results | Messages

	tid	coach	home_city
1	Lahore Lions	1 Mickey Arthur	Lahore
2	Karachi Kings	2 Wasim Akram	Karachi
3	Islamabad United	3 Dean Jones	Islamabad
4	Peshawar Zalmi	4 Darren Sammy	Peshawar
5	Sialkot Stallions	8 Moin Khan	Sialkot